Homework 2

(Due date: December 3, 2025)

This assignment investigates how monetary policy shocks propagate to global equity markets using an event study framework. Students will use Global Compustat country-level daily stock return data and the high-frequency FOMC monetary policy shock (NS shock) constructed by Nakamura and Swanson (2018, QJE) as the main measure of policy surprises. The shock data is provided on the class webpage.

Problem 1. Data Construction

Use daily country-level return data and Global market return benchmark (indexid = WORLD) from Compustat Global Index File. Merge these data with the high-frequency NS shock variable. Include all countries for which daily return data from the Compustat Global Index file are available throughout the full period covered by the NS shock dataset. Adjust for each country's time zone to define the 'first trading day after the announcement' (event day 0).

Problem 2. Event Study Design

Examine global stock market reactions over multiple post-announcement horizons. Compute abnormal and cumulative abnormal returns (CAR) for event day (0), and for +1, +2, +3, and +4 trading days. Estimate expected returns using the global market model:

$$R_{it} = \alpha_i + \beta_i R_{mt} + \varepsilon_{it},$$

where R_{mt} is the global market return from Compustat Global Index File (indexid=WORLD).

Problem 3. Regression Analysis on Monetary Policy Shocks

Estimate the following panel regression:

$$R_{it} = \alpha_i + \beta_1 \, NS\ Shock_t + \mu_i + \lambda_t + \varepsilon_{it},$$

where R_{it} is the cumulative event-window return and μ_i, λ_t represent country and event-date fixed effects.

Discuss the sign and magnitude of coefficients (β_1) and their economic interpretation.

(Bonus) Problem 4. Comparative Analysis

Extend your baseline event study regression by introducing one or more **country-level characteristics** that you find economically relevant (e.g., capital market openness, exchange-rate regime, financial development, or institutional quality). Specifically, estimate a model of the following form:

$$R_{i,t} = \alpha_i + \beta_1 NS\ Shock_t + \beta_2 Z_i + \beta_3 (NS\ Shock_t \times Z_i) + \mu_i + \lambda_t + \varepsilon_{i,t},$$

where Z_i represents a country-level variable that may shape the sensitivity of stock markets to monetary policy shocks.

Interpret how the country characteristic Z_i modifies the impact of NS shocks. Discuss possible economic mechanisms behind the heterogeneity in responses (e.g., capital mobility, dollar dependence, domestic monetary credibility, or policy synchronization).

Reference

Nakamura, E., & Swanson, E. T. (2018). High-Frequency Identification of Monetary Non-Neutrality: The Information Effect. Quarterly Journal of Economics, 133(3), 1283–1330.